

Errors reveal the depth of human reasoning

Tamal Maharaj ^{a,*}, Kenneth W. Regan ^b

^a*Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, West Bengal, India*

^b*Department of Computer Science and Engineering, University at Buffalo (SUNY), Buffalo, New York, USA*

Abstract

How deeply people think before choosing is the hidden quantity at the centre of bounded rationality, yet rarely measured in natural behaviour. We read it from the structure of human mistakes: across more than a million skill-graded chess decisions scored by an engine at every search depth, the moves of strong and weak players look comparably good to a shallow analysis and separate only as it deepens—what divides them is the depth at which their errors are exposed, the depth of satisficing. It rises with skill in slow play, contracts as the clock tightens, and vanishes in the fastest games. Estimated without timing information, it tracks how long players thought, recovers the known depth of synthetic agents, and improves move prediction only where deep reasoning has room to matter. Every central result survived a prospective replication registered before its data existed. What people get wrong, more than what they get right, reveals how they think.

Keywords: bounded rationality, satisficing, depth of reasoning, decision-making, expertise, chess

Author note.. Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is

*Corresponding author.
Preprint submitted to Cognition
Email addresses: tamal@gm.rkmvu.ac.in (Tamal Maharaj ) , regan@buffalo.edu (Kenneth W. Regan )
July 31, 2026

8 *split by player throughout. T. Maharaj formerly published as T. T. Biswas.*

9 **1. Introduction**

10 Sixty years of decision science rest on a quantity nobody can see. When
11 Simon proposed that people *satisfice*—search until an option clears an ac-
12 ceptability threshold, then stop—he made the depth of that search the load-
13 bearing variable of bounded rationality (Simon, 1955, 1956). The same hid-
14 den quantity does the work in the fast-and-frugal programme (Gigerenzer
15 and Goldstein, 1996), in resource-rational analysis, which treats cognition as
16 the optimal use of limited computation (Russell and Wefald, 1991; Griffiths
17 et al., 2015; Lieder and Griffiths, 2020; Callaway et al., 2022), in heuristics-
18 and-biases accounts of judgement (Tversky and Kahneman, 1974; Kahneman
19 and Tversky, 1979), and in the level- k and cognitive-hierarchy models of
20 strategic thinking (Stahl and Wilson, 1995; Nagel, 1995; Costa-Gomes et al.,
21 2001; Camerer et al., 2004; Costa-Gomes and Crawford, 2006; Crawford et al.,
22 2013; Strzalecki, 2014). Yet in practice, how far a person looks before com-
23 mitting has been inferred almost entirely from small staged experiments.
24 A notable recent exception used a controlled board game with behavioural
25 modelling to show that planning depth grows with expertise (van Opheusden
26 et al., 2023). Whether such regularities hold in real, repeated, consequential
27 choice—outside the laboratory and at the scale of populations—has been
28 hard even to ask, because natural decisions rarely come with a ground truth
29 for how good each option was, let alone for how good it *looked* at each suc-
30 cessive depth of consideration.

31 Chess supplies exactly that, which is why it has served for a century as

32 the model organism of expert cognition (de Groot, 1965; Newell and Simon,
33 1972; Chase and Simon, 1973; Gobet and Simon, 1996; Ericsson et al., 1993).
34 Millions of decisions by players of precisely known skill are recorded in real
35 competition, and every one of them can be scored by an engine far stronger
36 than any human (Campbell et al., 2002; Silver et al., 2018)—not once, but
37 at every depth of an iterative-deepening search. The engine’s opinion of a
38 move changes as it looks deeper: a move that seems strong at a glance may
39 be refuted only far down the tree, and a move that looks dull at first may
40 reveal its point three moves later. This depth-resolved record lets us put a
41 new question to every human mistake: *at what depth does the flaw become*
42 *visible?* And therefore: how deep must the player have looked?

43 We call the answer the **depth of satisficing**—the depth at which the
44 move a player chose stops looking better than the move they should have
45 played. Our central empirical claim is that this is the dimension along which
46 skill is most legible. Judged only at shallow depth, the moves of players
47 separated by a thousand rating points look comparably good; the gap opens
48 as the engine searches deeper. What divides them is carried by their errors—
49 not by how many they make, but by how deep one must look before the flaw
50 appears. Skill is present even in moves played at a glance, since stronger
51 players recognise better options without searching; but most of what rating
52 buys is realised only through the depth to which a player effectively searches.

53 Online chess also supplies a natural, graded manipulation of how deeply
54 a person can afford to look: the time control. A player with an hour for
55 the game can follow a line many moves ahead; one with ten minutes cannot;
56 and one playing *bullet*—a minute or two for all the moves—has time only

57 to react. Because the same players compete across every control, the clock
58 varies the budget for search while holding the person, and their pattern
59 knowledge, fixed. If the depth of satisficing measures search rather than
60 mere skill, it should track that budget: deepest in classical play, shallower
61 in rapid and blitz, and collapsing in bullet, where even strong players must
62 move on recognition alone with no time to verify what they see. This graded
63 prediction across four controls is among the sharpest tests we can put to
64 the construct, and its limiting case is a built-in control: in bullet, search is
65 unavailable to everyone, so any residual skill difference there is pattern, not
66 depth.

67 Three things distinguish this work from earlier studies of chess errors and
68 engine depth (Regan and Haworth, 2011; Biswas and Regan, 2015; McIlroy-
69 Young et al., 2020). First, we replace descriptive curve-fitting with a genera-
70 tive model: the player is a latent distribution over search depths, so the depth
71 of satisficing is a posterior quantity with stated uncertainty rather than the
72 crossing point of two fitted curves. Second, we do not assume that engine
73 depth is a valid proxy for human reasoning depth; we test it—predictively,
74 convergently, and differentially. Third, we subjected the construct to an
75 adversarial replication programme: hypotheses, pipeline, and decision rules
76 were pre-registered and then confirmed on untouched months of data, ending
77 with a fully prospective test registered before its data existed (Nosek et al.,
78 2018). The final, registered confirmation succeeded on all three primary
79 hypotheses.

2. Results

Reading a decision at increasing depth.. A chess engine evaluates by *iterative deepening*: it scores every candidate move looking one ply (half-move) ahead, then two, and so on, revising each move’s evaluation at every new horizon; evaluations change with depth because deeper search uncovers consequences that shallower search cannot see—the game-tree analogue of thinking further ahead. A move’s value is therefore not a number but a *trajectory* across depth, and that trajectory is our raw material. Figure 1a shows a real decision from the data. A 1910-rated player, offered a bishop, recaptured it (fxe6): at shallow depth the capture looks close to winning, but from about eight plies the refutation comes into view and its value collapses. The quiet king move Kb8 shows the opposite signature— unimpressive at a glance, it proves best once the search runs deep. Writing $\delta_{m,d}$ for move m ’s *regret* at depth d —the win probability it concedes relative to the best option visible at that depth (Methods)—we call the signed area between a move’s regret trajectory and its full-depth endpoint the move’s *swing*, $\text{sw}(m) = \sum_d (\delta_{m,d} - \delta_{m,D})$: positive (*swing-up*, a deep discovery) when a move looks worse shallow than it truly is, negative (*swing-down*, a trap) when shallow search flatters it. A position’s total swing, $\sum_m |\text{sw}(m)|$, measures how much the picture changes with depth—how much deliberation the position can reward. The depth at which the played move’s flaw or worth becomes visible is the signal everything below is built on.

Skill lives in the errors, not the successes.. Our primary sample comprises 922,412 decisions by 16,328 players rated 1200 to 2600+, drawn from online play and from elite over-the-board tournaments (Methods). At shallow en-

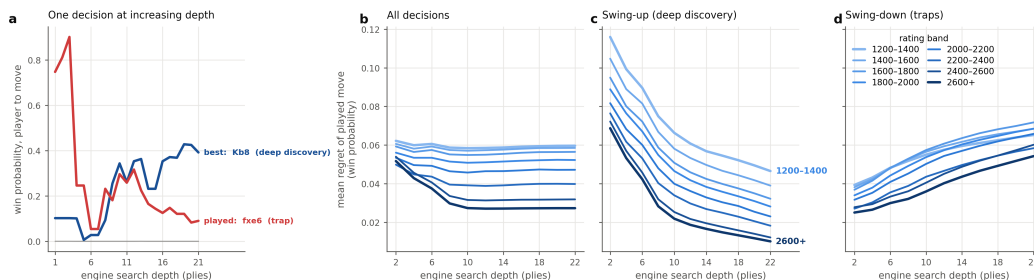


Figure 1: **Skill separates only at depth, and mostly on deep-discovery moves.**

a, A worked example: candidate-move win probabilities across engine search depth for one real decision (grey, other candidates; moves in standard algebraic notation). The played pawn capture *fxe6* looks nearly winning at shallow depth and collapses once the refutation surfaces (a *trap*, swing-down); the quiet king move *Kb8* reveals its worth only deep (a *deep discovery*, swing-up). **b–d**, Mean regret of the played move across engine search depth, by rating band (light to dark blue: 1200–1400 up to 2600+), within the slow controls (classical and rapid). **b**, All decisions: rating bands are nearly indistinguishable at shallow depth and fan apart as the engine looks deeper. **c**, Swing-up (deep-discovery) decisions: a clean, well-ordered rating fan—the rating signal concentrates here. **d**, Swing-down (trap) decisions: bands bunch and overlap, separating roughly half as much. The rating axis spans online play (lower bands) to elite over-the-board play (upper bands).

105 gine depth, the average error of the played move barely distinguishes a club
 106 player from a grandmaster; the gap opens only as the engine looks deeper
 107 (Fig. 1b). Because online ratings are pool-specific—a separate rating per
 108 time control—we read every skill comparison *within* a control: at full depth,
 109 mean regret of the played move falls with rating in classical (0.059 at the
 110 weakest band to 0.028 at the strongest), in rapid, and in blitz, but stays
 111 nearly flat in bullet (0.070 to 0.055), where no one has time to search deeply.
 112 The separating information, moreover, is not spread evenly across mistakes.
 113 It is concentrated in *deep-discovery* decisions—swing-up moves whose worth

only emerges at depth—which fan out cleanly by rating (Fig. 1c), while trap-like swing-down decisions bunch the rating bands together (Fig. 1d). A single swing-up decision carries roughly twice the rating information of a swing-down one (pool-normalised single-decision $R^2 \approx 0.006$ versus 0.003), an asymmetry that recurs in every replication month below. What matters is not how large an error is, but at what depth the move’s true worth appears.

Depth of satisficing rises with skill and collapses under time pressure.. Fitting the latent-depth model (Methods) and reading off each decision’s posterior expected depth, the construct behaves exactly as a measure of deliberation should—provided, again, that one looks within a time control, since pooling would conflate skill with available clock. Within classical games, the depth of satisficing climbs from about 10.8 plies in the lowest rating band to about 13.7 in the highest, with non-overlapping player-clustered 95% intervals at the extremes (Spearman $\rho = +0.40$; rapid +0.66; blitz +0.45; Fig. 2a). Bullet is the built-in negative control: the relationship vanishes ($\rho = -0.04$, flat across bands). When nobody can search, depth stops separating anybody, which rules out the construct being a re-description of rating. The same signature appears *within* players: de-meaned depth rises across think-time quartiles, from -0.14 plies on a player’s fastest quarter of moves to $+0.33$ on their slowest (Fig. 2b).

Inferred depth tracks real thinking time it was never shown.. The construct’s key non-circular test removes the clock from the model entirely. Refit with the clock feature zeroed, so that timing can play no role in the estimate, the model still assigns held-out depths that correlate with how long the player actually spent on the move (Spearman $\rho = +0.36$ overall), and the

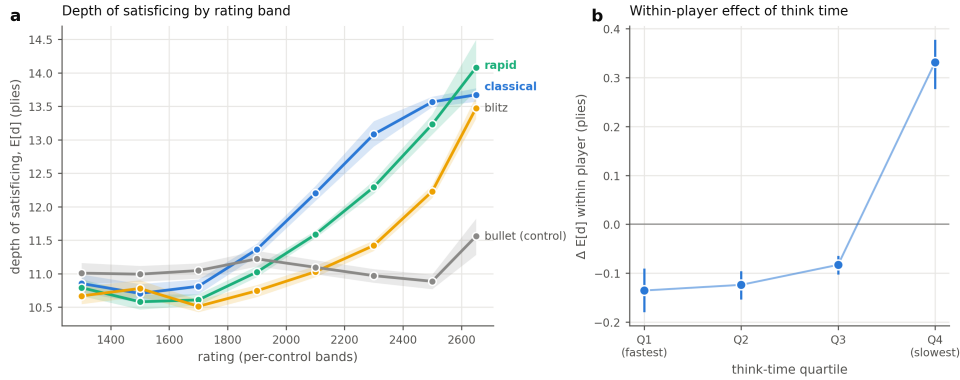


Figure 2: **Depth of satisficing rises with skill and with time to think.** **a**, Posterior depth of satisficing by rating band, within each time control (shaded bands, player-clustered bootstrap 95% CIs). Depth rises with rating in classical (Spearman $\rho = +0.40$), rapid (+0.66) and blitz (+0.45); bullet, where no one can search, is the flat negative control ($\rho = -0.04$). **b**, Within-player effect: de-meaned depth by think-time quartile (dots, pooled means; bars, player-clustered bootstrap 95% CIs). The same players search deeper on their slower moves.

139 correlation peaks in the middlegame (+0.40)—the phase where deliberation
 140 genuinely varies most—falling to +0.33 in the opening and +0.29 in the
 141 endgame (Fig. 3). Inferred depth recovers observed deliberation without
 142 ever being told about time. That is convergent evidence that the posterior
 143 reflects thinking, not the rating prior it partly leans on.

144 *A depth-aware model predicts human moves better, exactly where depth should*
 145 *matter..* If the depth of satisficing is real, knowing it should help predict
 146 what a human will play, and the help should be confined to situations where
 147 depth can vary and can matter. Both hold. A fusion of a state-only human-
 148 move model (Maia-3 (Monroe et al., 2026)) with a mixture-over-depths search
 149 model improves held-out cross-entropy over the state-only model by a modest

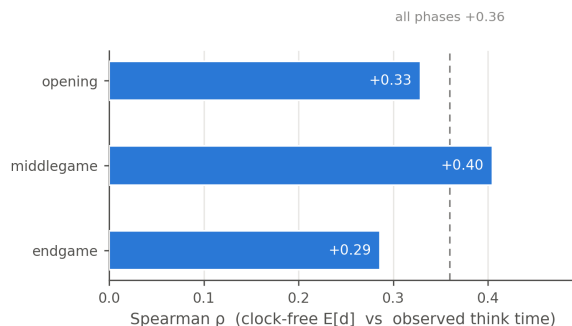


Figure 3: **Depth inferred without the clock tracks real thinking time.** Held-out Spearman correlation between the clock-free model’s per-decision depth and observed time-on-move, by game phase (dashed line, all phases pooled). The correlation peaks in the middlegame, where deliberation varies most.

0.012 nats overall (95% CI [0.008, 0.016], player-clustered bootstrap; top-1 accuracy +0.7 points). The scientific content is not the average but the pre-registered differential: the gain concentrates in classical, high-swing positions (+0.053 nats) and is absent or negative in fast or low-swing play (Fig. 4). The swing-magnitude $\times$ time-control interaction is +0.060 nats (95% CI [0.056, 0.064]), confirmed by a mixed-effects model with players as random effects ($\beta = 0.058$, $p < 10^{-277}$). No state-only model can produce this pattern, however strong (Ruoss et al., 2024); within classical play the gain falls almost entirely on swing-up moves (+0.048 nats)—precisely the choices whose worth a pattern model cannot see because it does not look ahead.

Positions are test items; swing is their discrimination.. These findings admit a measurement-theoretic reading that unifies them. Treat each position as a test item whose *difficulty* is its critical depth—the deepest ply at which a misleading move still looks best—and whose graded *response* is the severity of the move played (best, inaccuracy, mistake, blunder). An explanatory

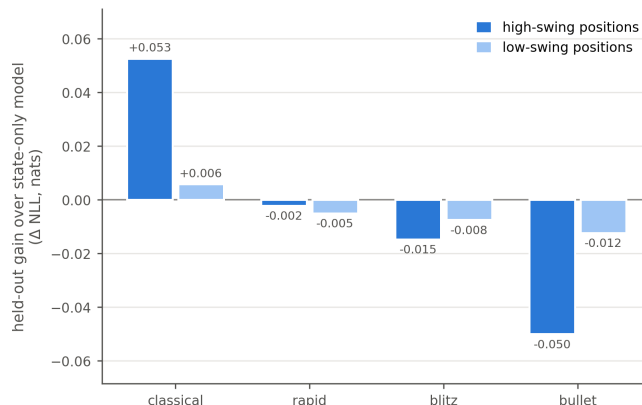


Figure 4: **Depth improves move prediction only where depth can matter.** Held-out gain of the depth-aware fusion over the state-only model (nats), by time control and position swing. The gain concentrates in classical, high-swing positions and is absent or negative in fast or low-swing play—the pre-registered differential.

165 graded-response model (Samejima, 1969), with item properties entered as
 166 engine-derived covariates in the manner of a linear logistic test model (Fis-
 167 cher, 1973) (each natural position occurs only once, so per-item parameters
 168 are unidentifiable), recovers the full expected structure: error severity falls
 169 with ability ($\beta = -0.13$, $p < 10^{-80}$), rises with item difficulty ($\beta = +0.06$,
 170 $p < 10^{-24}$), and—the decisive term—ability discriminates more sharply in
 171 high-swing positions (ability $\times$ swing $\beta = -0.08$, $p < 10^{-38}$; Fig. 5a). In
 172 item-response language, *swing is item discrimination*: test information con-
 173 centrates in high-swing decisions, which is exactly why skill is legible in
 174 so small a fraction of moves. The graded operating curves shift cleanly
 175 along the ability axis (Fig. 5b), and item difficulty lives on the same ply
 176 ruler as the person-side depth of satisficing (Fig. 5c)—a natural generalisa-
 177 tion of rating systems, which are themselves one-parameter item-response



Figure 5: **An item-response reading: swing is item discrimination.** **a**, Probability that the played move is not the engine-best, versus ability (within-pool rating z), by position-swing tercile: ability discriminates far more steeply in high-swing positions. **b**, Graded-response operating curves: probability of each severity category versus ability; $P(\text{best})$ rises and $P(\text{blunder})$ falls monotonically. **c**, Items placed on the ply ruler by their critical depth—the same scale on which the depth of satisficing measures person ability.

models on game outcomes (Elo, 1978; Rasch, 1960; Embretson and Reise, 2000). The structure is not a platform artefact. On an independent pool of 99,358 chess.com decisions by 945 players—a different site, a different rating system—the same model reproduces every sign (ability $\beta = -0.08$, difficulty $+0.11$, swing $+0.87$, ability $\times$ swing -0.06 ; all $p < 10^{-13}$), and the same holds on a second Lichess month (2026-04: ability $\times$ swing $\beta = -0.09$, $p < 10^{-40}$)—three independent pools in all. At the present decision density per player, ability is reliable only in aggregate; person-dense, pool-invariant ability estimation is future work.

Profiles, not a single number, separate expert styles.. Averaged per player from an *elo-free* refit (so that recovering rating is not circular), four behavioural axes—depth of satisficing, trap susceptibility, deep-discovery rate, and time elasticity—summarise a player’s style. Across 873 players with

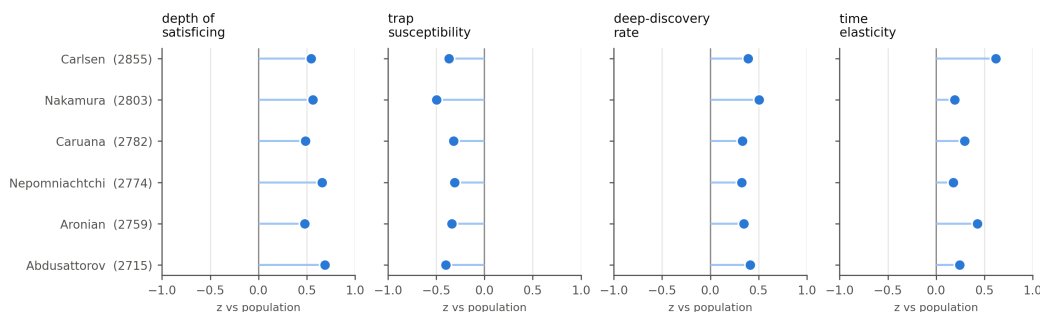


Figure 6: **Four behavioural axes separate elite styles that one number confuses.**

Profile z-scores (versus the full population) for the six elite players with the most analysed decisions, ordered by rating. A one-dimensional depth score inverts elite ratings (9 of 15 pairs discordant); the four-axis profile recovers within-pool rating across 873 players ($R^2 = 0.10$, nested cross-validation) where depth alone recovers none ($R^2 \approx 0$), and reads the styles directly—for example Nakamura as most trap-resistant but among the least time-elastic, Carlsen as converting think time into depth most steeply.

191 at least 100 decisions, the four-axis profile recovers within-pool rating at
192 $R^2 = 0.10$ under nested cross-validation while the depth axis alone recovers
193 none ($R^2 \approx 0$): the added axes carry the signal, reported on the honest
194 within-pool scale rather than the cross-pool one (which flatters recovery by
195 conflating skill with pool membership). Among elite players the profile also
196 resolves a known embarrassment of one-dimensional scores: a bare depth
197 ranking inverts the true order of the world’s best (9 of 15 elite pairs dis-
198 cordant), ranking, for example, Abdusattorov and Nepomniachtchi above
199 Carlsen. The four axes instead attribute each player’s strength differently—
200 Nakamura is the most trap-resistant of the six yet among the least time-
201 elastic (a blitz specialist who reaches his depth *fast*), while Carlsen converts
202 thinking time into depth more steeply than anyone—and the inversions dis-
203 appear (Fig. 6).

204 *The instrument recovers depth it has never seen..* Synthetic agents that satis-
 205 fice at fixed, known depths, played through the identical pipeline, come back
 206 in the right order: recovered depth increases monotonically with planted
 207 depth (group-level Spearman = 1.0 across planted depths 4–16), and the es-
 208 timator independently recovers the agents’ decision sharpness ($\hat{\beta} = 7$ against
 209 a generating $\beta = 6$; Fig. 7). Recovery is ordinal rather than metric—per-
 210 agent estimates are noisy and compressed toward the middle (mean absolute
 211 error 3.7 plies on roughly 70 decisions per agent)—but it establishes that the
 212 model measures depth rather than position difficulty. Recovery requires a
 213 flat prior over depth: imposing the human rating-conditioned prior collapses
 214 the estimate toward its mean, which is why the two non-circular tests above,
 215 and not the posterior’s absolute scale, carry the evidential weight.

216 *A pre-registered programme ending in a prospective confirmation..* We pre-
 217 registered hypotheses, pipeline, and decision rules, and then ran the frozen
 218 pipeline once on each of three untouched Lichess months (Nosek et al., 2018).
 219 The first replication (2026-05; 179,360 decisions; OSF KB4ZQ) confirmed
 220 the non-circular core—clock-free depth tracked thinking time ($\rho = +0.30$,
 221 middlegame peak), and the registered swing $\times$ time-control interaction held
 222 ($+0.049$ nats, mixed-effects $p < 10^{-36}$)—but failed its own strict rule because
 223 one arm of one primary (blitz, registered as positive) came out flat, and it
 224 exposed a swing-label convention error: the single-decision rating signal sits
 225 in swing-*up* decisions, not swing-*down* as first registered. We disclosed both,
 226 corrected the hypotheses, and registered a second, fully prospective protocol
 227 (OSF A6NYK) whose primary dataset—Lichess 2026-06—did not yet exist
 228 at registration. A robustness month processed under the same registration

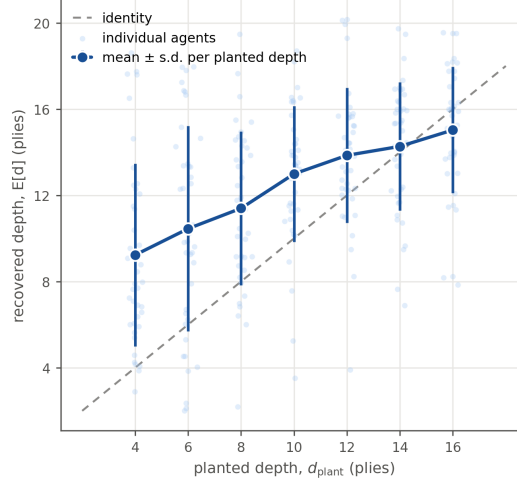


Figure 7: **The instrument recovers planted search depth.** Synthetic agents satisficing at fixed depths (4–16 plies), run through the identical pipeline. Pale dots, individual agents; dark points, mean $\pm$ s.d. per planted depth; dashed line, identity. Recovery is monotone (group-level Spearman = 1.0) though ordinal: per-agent estimates are compressed toward the middle (MAE 3.7 plies at ~ 70 decisions per agent).

229 (2026-04; 168,654 decisions) meanwhile reproduced the slow-control rating
 230 relationship (classical $\rho = +0.47$, rapid $+0.54$), the convergent-time result
 231 ($+0.40$), and the interaction ($+0.076$ nats, $p < 10^{-48}$).

232 The prospective month then arrived and was processed once. All three
 233 primary hypotheses replicated, satisfying the registered decision rule. On
 234 203,248 decisions from 3,707 players: depth rose with rating within ev-
 235 ery slow control (classical $\rho = +0.23$, 95% CI $[+0.15, +0.35]$; rapid $+0.72$
 236 $[+0.59, +0.79]$) with bullet null as predicted ($+0.08 [-0.03, +0.19]$); clock-
 237 free depth tracked observed thinking time ($\rho = +0.32$, middlegame peak
 238 $+0.36$); and the depth-aware model beat the state-only model overall ($+0.010$
 239 nats, 95% CI $[+0.003, +0.017]$) with the registered interaction intact ($+0.068$

240 nats $[+0.055, +0.079]$; mixed-effects $\beta = +0.059$, $p < 10^{-42}$). Two sec-
 241 ondary outcomes are worth reporting straight. Blitz, which we had over-
 242 corrected to “null” after its flat 2026-05 showing, came back clearly positive
 243 ($\rho = +0.39 [+0.25, +0.50]$); across the four months the blitz relationship now
 244 reads $+0.45, \approx 0, +0.19, +0.39$ —real but unstable, in contrast to the robust
 245 slow controls and the reliably null bullet. And the deep-discovery asymmetry
 246 replicated in its corrected direction (swing-up $R^2 = 0.0049$ versus swing-down
 247 0.0027), as it has in every month examined. The construct’s central claims
 248 thus rest on a confirmation that was, in the strict sense, predicted in public
 249 before the behaviour occurred.

250 **3. Discussion**

251 Errors are usually the noise of behavioural data. Here they are the sig-
 252 nal. Where success is widely shared, the fine structure of failure—the depth
 253 at which a flaw becomes visible—turns out to be a stable, measurable sig-
 254 nature of how a person reasons: it scales with expertise, contracts under
 255 time pressure, tracks deliberation it was never shown, and improves pre-
 256 diction exactly in the regime where deep reasoning has room to operate.
 257 The pattern supports a specific mechanism: people commit to the best op-
 258 tion found at the search horizon they reach, rather than integrating over
 259 the whole trajectory—consistent with theoretical results that level- k predic-
 260 tions are independent of higher-order belief tails (Strzalecki, 2014), and con-
 261 necting satisficing to model-based control (Daw et al., 2005), decision-tree
 262 pruning (Huys et al., 2012), and bounded evidence accumulation (Ratcliff
 263 and McKoon, 2008). Where planning depth has been measured before, it

264 required controlled laboratory games and explicit search models (Callaway
265 et al., 2022; van Opheusden et al., 2023); here the same construct is read
266 non-invasively from natural, high-stakes play at population scale, against a
267 depth-resolved value standard, and it survives a prospective registered test.
268 The item-response formulation adds a measurement-theoretic dividend: po-
269 sitions become items with engine-derived difficulty and discrimination, so
270 people and problems share one ply-denominated scale—rating systems gen-
271 eralised from game outcomes to individual decisions.

272 The individual profiles suggest what the construct is for. Expertise is
273 not a scalar: two players of equal rating can owe their strength to depth,
274 to trap resistance, to deep discovery, or to the efficiency with which time
275 becomes depth—echoing evidence that expert flexibility itself varies (Bilalić
276 et al., 2008). An agent that estimates a person’s depth of satisficing could,
277 in principle, select problems pitched just beyond it; a rating system built
278 on decisions rather than outcomes could equate skill across pools that never
279 play each other.

280 The limitations are the flip side of the design. Engine search depth is a
281 *validated proxy* for human selective search—validated by convergence with
282 thinking time and by recovery of planted depth—but it is not isomorphic to
283 human search, which is narrow, uneven, and pattern-guided. The depth-of-
284 satisficing posterior leans partly on a rating-conditioned prior; per-decision
285 move evidence is weak, which is why the clock-free and synthetic tests, not
286 the posterior’s absolute scale, carry the claims. The average predictive gain
287 is small, and the case rests on its registered differential rather than its size.
288 The blitz rating–depth relationship is genuinely variable across months and

289 should not be quoted as a stable effect. And chess players are a highly
290 trained, self-selected population in an adversarial task; the approach extends
291 in principle to any domain with skill-graded decisions and a depth-resolved
292 value standard—a value network queried at increasing depth in Go, or staged
293 value estimates in planning tasks—but that generality remains to be demon-
294 strated.

295 4. Methods

296 *Data..* The primary dataset comprises 922,412 decisions by 16,328 players:
297 online games from the Lichess open database (month 2025-09), balanced-
298 sampled across rating bands (200-point bins, 1200–2600+) and time classes
299 (bullet, blitz, rapid, classical), with per-move clock times retained. Time
300 classes follow Lichess’s convention, by estimated game duration (initial time
301 plus $40\times$ the increment): bullet under 3 minutes, blitz 3–8, rapid 8–25, and
302 classical 25 minutes or more. Each class therefore pools nearby formats—
303 1+0 and 1+1 bullet, for instance, both fall in the bullet class despite an ap-
304 preciable difference in effective time—so residual within-class heterogeneity
305 remains, and every skill comparison in this paper is made within class, never
306 across; plus an elite over-the-board stratum of 371,279 decisions from 1,261
307 titled players in broadcast tournament games, which supports the individual-
308 profile case study. Replication datasets—Lichess 2026-05 (179,360 decisions),
309 2026-04 (168,654), and the prospective 2026-06 (203,248 decisions, 3,707
310 players, 113,421 classical)—were each sampled by the same frozen pipeline
311 (identical seed, cell targets, and engine settings) and processed once. The
312 independent-pool item-response check uses 99,358 decisions by 945 titled

313 chess.com players harvested via the public Published-Data API. Synthetic
 314 agents (below) provide ground truth. All decisions are restricted to plies
 315 9–120 to skip opening-book play.

316 *Engine analysis and value scaling.* Positions are analysed with Stockfish 18
 317 in multi-PV mode, recording each candidate move m_i ’s evaluation at every
 318 depth $d = 1 \dots D$ ($D=21$, $K=9$). The candidate set is the union of the top- K
 319 across all depths, so trap moves keep a full trajectory. Because consecutive
 320 plies of each game are analysed, the value of a played move falling outside the
 321 top- K is read off from the position it leads to (the next analysed decision):
 322 its win probability at depth d equals one minus the successor’s best win
 323 probability at depth $d-1$, matching the other candidates’ negamax depth.
 324 Where no analysed successor exists (game end, final sampled ply) the played
 325 move takes the weakest retained candidate’s win probability at each depth.
 326 With `UCI_ShowWDL` the engine reports win/draw/loss counts (w, dr, ℓ) per
 327 mille, giving the win probability $P_{i,d} = (w + \frac{1}{2} dr)/1000$ directly—no logistic
 328 or centipawn scaling. Depth- d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \geq 0. \quad (1)$$

329 Being a difference against the best move *at each depth*, δ is invariant to
 330 anything that shifts all moves together (position difficulty, evaluation scale),
 331 isolating the choice problem. Searches run to depth D or a 2×10^8 -node
 332 cap, whichever comes first; the 0.09% of positions that stop short have their
 333 unreached depths masked in the likelihood (reached depth recorded) rather
 334 than imputed.

335 *Pattern prior.* The human-pattern prior $q_i(p, c)$ is the Maia-3 policy (Mon-
 336 roe et al., 2026) at the side-to-move’s rating, read from the policy head as a
 337 full distribution over legal moves.

338 *Latent-depth choice model.* A player who has effectively searched to depth
 339 d chooses by a soft-max over depth- d regret fused with the pattern prior as
 340 a product of experts,

$$P(\text{move}=i \mid d, p, c) = \frac{\exp(-\beta \delta_{i,d} + \alpha \log q_i)}{\sum_j \exp(-\beta \delta_{j,d} + \alpha \log q_j)}. \quad (2)$$

341 Effective depth is latent with distribution $\pi_d(p, c)$, emitted by a small net-
 342 work over a coarse depth grid conditioned on context c (rating, time class,
 343 clock, move number, total swing). The move likelihood marginalises depth,
 344 $P(\text{move}=i \mid p, c) = \sum_d \pi_d(p, c) P(\text{move}=i \mid d, p, c)$, and parameters minimise
 345 the negative log-likelihood of the played move with an entropy penalty on π
 346 for identifiability and β tied globally. Data are split by player throughout.

347 *Depth of satisficing.* The posterior over effective depth given the observed
 348 move,

$$r_d = P(d \mid y, p, c) = \frac{\pi_d P(y \mid d)}{\sum_{d'} \pi_{d'} P(y \mid d')}, \quad \hat{d} = \sum_d d r_d, \quad (3)$$

349 gives a per-decision depth with uncertainty. Lichess ratings are pool-specific
 350 (a separate Glicko-2 rating per time control, not comparable across controls),
 351 so no analysis pools ratings across controls; doing so would also confound skill
 352 with available clock. Per-band estimates carry player-clustered bootstrap
 353 95% intervals (1,000 resamples of players) (Efron, 1979).

354 *Relation to the curve-intersection estimator.* The construct originates in
 355 a population-level heuristic (Biswas and Regan, 2015): average the played

356 move’s error and the engine move’s error against depth, and read the depth
 357 of satisficing off the curves’ crossing. That estimator returns one depth per
 358 cohort, carries no per-decision uncertainty, presupposes a single time control,
 359 and inherits the fit (cohort depths move by roughly a ply under reasonable
 360 fitting choices). We keep the underlying signal—a played move whose error
 361 grows with depth is exactly the move *swing* of Biswas and Regan (Biswas
 362 and Regan, 2015)—and replace the crossing with the generative posterior r_d :
 363 a per-decision quantity with credible intervals that supports within-control
 364 and within-player contrasts and is identifiable against synthetic ground truth.
 365 The population curve is stable only because it averages thousands of weak
 366 per-move signals, which the crossing view conceals and our identifiability
 367 test makes explicit: recovered depth is ordinal and prior-compressed, which
 368 is why the convergent and identifiability tests, not the posterior’s absolute
 369 scale, carry the claims. Where the two views can be compared they agree—
 370 stronger players satisfice deeper and err less—now within time control, at far
 371 larger scale, on win-probability evaluation, with a human-pattern prior, and
 372 under a registered replication programme.

373 *Item-response formulation..* Each decision is an item. Difficulty b_j is the
 374 deepest grid depth at which the apparent-best move differs from the full-
 375 depth best move—the critical depth d_i of Biswas and Regan (Biswas and
 376 Regan, 2015), in plies; the ordered response is the played move’s full-depth re-
 377 gret binned as best (≤ 0.02), inaccuracy (≤ 0.05), mistake (≤ 0.10), or blun-
 378 der (> 0.10) in win-probability units. We fit an explanatory proportional-
 379 odds graded-response model (Samejima, 1969; Fischer, 1973) with ability
 380 (within-pool rating z), b_j , position swing, and ability $\times$ swing as predictors;

381 item properties enter as covariates rather than free parameters because most
382 natural positions occur exactly once. The MLE uses a random subsample;
383 response curves are computed on all decisions. Per-player ability has low
384 split-half reliability at this decision density, so effects are reported at the
385 item and rating-band level.

386 *Profiles and rating recovery..* Per player with at least 100 decisions: mean
387 $\hat{d}$; trap susceptibility (rate of played swing-down moves, weighted by shallow
388 attractiveness); deep-discovery rate (rate of played swing-up moves); and
389 time elasticity (within-player slope of $\hat{d}$ on log think-time). The depth axis
390 comes from a model fit with the rating feature removed, keeping recovery non-
391 circular. Rating is predicted from the standardised profile by ridge regression
392 under nested five-fold cross-validation (inner folds select the penalty), within
393 pool (z-scored per time control, over-the-board play its own pool). The elite
394 case study reports the standardised axes for the six broadcast players with
395 the most analysed decisions.

396 *Model comparison and the registered differential..* State-only ($\beta=0$), search-
397 only ($\alpha=0$), and fusion predictors are fit under one player split and compared
398 on held-out cross-entropy and top-1 match. The registered hypothesis is an
399 interaction: the fusion’s gain over state-only is larger in high-swing than low-
400 swing positions and in classical than blitz. It is tested with a player-clustered
401 bootstrap and a linear mixed-effects model with player random intercepts.

402 *Convergent and recovery validation..* Convergent: refit with the clock feature
403 zeroed, then correlate held-out $\hat{d}$ with observed time-on-move, by game phase.
404 Recovery: agents play a soft-max ($\beta_{\text{gen}}=6$) over fixed-depth win probabilities;

405 their games pass through the identical pipeline, and a per-agent flat-prior
406 posterior over depth (search likelihood, with the population β fit) is compared
407 with the planted depth.

408 *Pre-registered replication programme..* Hypotheses, pipeline, sampling seed,
409 and the decision rule (success iff all three primary hypotheses replicate di-
410 rectionally with stated intervals) were registered before analysing each repli-
411 cation month (first registration: OSF KB4ZQ; corrected and prospective
412 second registration: OSF A6NYK, whose primary month 2026-06 was regis-
413 tered before Lichess published the data). Each month was processed exactly
414 once by the frozen pipeline. Deviations are disclosed in Results: the first
415 registration’s blitz prediction was too strong and its swing-direction conven-
416 tion reversed; the second registration’s blitz-null prediction proved too weak
417 in turn (blitz positive in the prospective month). Neither deviation touches
418 the primary hypotheses.

419 *Reproducibility..* Stockfish 18 (shared NNUE), Threads 1 per process across
420 many single-thread engines, Hash 256 MB, $D=21$, $K=9$, 2×10^8 -node cap,
421 driven through the `python-chess` UCI bridge (v1.11), which also provides all
422 board and move representation; Maia-3 (`maia3-79m`) ([Monroe et al., 2026](#));
423 depth grid $\{2, 4, \dots, 22\}$; sampling seed 17 and 60 games per (rating band $\times$
424 time class) cell in every month; train/test split by player; model fits seeded.
425 All analyses ran on a single dual-socket 64-core server with one GPU; the
426 engine stage dominates the compute at roughly 1,400 core-hours per month
427 of data, and all model fitting and analysis complete within about an hour.
428 Strata and the primary interaction were fixed before held-out evaluation;
429 post-hoc choices are labelled exploratory.

430 *Ethics..* This study is a secondary analysis of publicly available, pseudony-
431 mous game records; no participants were recruited, no interaction or inter-
432 vention took place, and no private or identifiable personal information was
433 collected or processed. Online players appear only under their self-chosen
434 public usernames; the named elite players analysed in the individual profiles
435 are public figures whose over-the-board tournament games are published as
436 a matter of course. As research limited to secondary analysis of public data,
437 the study did not require prior ethics approval and no formal ethics review
438 was sought; informed consent was not required for the same reason.

439 *Data availability..* All game records derive from public sources: the Lichess
440 open database (<https://database.lichess.org>; months 2025-09, 2026-04,
441 2026-05, 2026-06), openly licensed Lichess broadcast exports of over-the-
442 board play, and public games retrieved via the chess.com Published-Data API
443 (<https://api.chess.com>). The derived analysis datasets—per-position en-
444 gine depth trajectories, Maia-3 policies, and the model tensors behind every
445 result, for the primary month and all three replication months—are deposited
446 in the study’s Open Science Framework project (<https://osf.io/ruhy8>)
447 under a CC-BY licence; the deposit DOI will be minted on publication.

448 *Code availability..* The full pipeline (engine driver, dataset builder, model,
449 confirmatory replication script, and all figure code) and configuration files are
450 openly available under the MIT licence at [https://github.com/tamalrkm/](https://github.com/tamalrkm/depth-of-satisficing)
451 [depth-of-satisficing](https://github.com/tamalrkm/depth-of-satisficing). The pre-registrations are archived in the same OSF
452 project (registrations [10.17605/OSF.IO/KB4ZQ](https://osf.io/10.17605/OSF.IO/KB4ZQ) and [10.17605/OSF.IO/A6NYK](https://osf.io/10.17605/OSF.IO/A6NYK)).
453 Stockfish 18 and Maia-3 are third-party open-source software.

454 *Acknowledgements..* The authors gratefully acknowledge the support from
455 the Department of Science and Technology (DST), Government of India,
456 through the DST-FIST grant [Sanction No.: SR/FST/MS-I/2022/116].

457 *Author contributions..* T.M. conceived the study, designed and implemented
458 the latent-depth model and the full analysis pipeline, ran the experiments,
459 pre-registered and ran the replication programme, and drafted the manuscript.
460 K.W.R. contributed the intrinsic-performance-rating and move-swing frame-
461 work underlying the construct, advised on chess-cognition methodology and
462 validation design, and critically reviewed the manuscript. Both authors ap-
463 proved the final version.

464 *Competing interests..* The authors declare no competing interests.

465 *Ethics..* The study uses only publicly released, openly licensed records of
466 voluntary online and broadcast play; it involves no intervention and no pri-
467 vate or personally identifying data beyond the public usernames and player
468 names attached to those records, and so did not require ethical approval.

469 **References**

- 470 Bilalić, M., McLeod, P., Gobet, F., 2008. Inflexibility of experts—Reality
471 or myth? Quantifying the Einstellung effect in chess masters. *Cognitive*
472 *Psychology* 56, 73–102.
- 473 Biswas, T.T., Regan, K.W., 2015. Measuring level-k reasoning, satisficing,
474 and human error in game-play data, in: 2015 IEEE 14th International
475 Conference on Machine Learning and Applications (ICMLA), pp. 941–947.
476 doi:[10.1109/ICMLA.2015.233](https://doi.org/10.1109/ICMLA.2015.233).

- 477 Callaway, F., van Opheusden, B., Gul, S., Das, P., Krueger, P.M., Grif-
478 fiths, T.L., Lieder, F., 2022. Rational use of cognitive resources in hu-
479 man planning. *Nature Human Behaviour* 6, 1112–1125. doi:[10.1038/
480 s41562-022-01332-8](https://doi.org/10.1038/s41562-022-01332-8).
- 481 Camerer, C.F., Ho, T.H., Chong, J.K., 2004. A cognitive hierarchy model of
482 games. *The Quarterly Journal of Economics* 119, 861–898.
- 483 Campbell, M., Hoane Jr., A.J., Hsu, F.h., 2002. Deep Blue. *Artificial Intel-
484 ligence* 134, 57–83.
- 485 Chase, W.G., Simon, H.A., 1973. Perception in chess. *Cognitive Psychology*
486 4, 55–81.
- 487 Costa-Gomes, M., Crawford, V.P., Broseta, B., 2001. Cognition and behavior
488 in normal-form games: An experimental study. *Econometrica* 69, 1193–
489 1235.
- 490 Costa-Gomes, M.A., Crawford, V.P., 2006. Cognition and behavior in two-
491 person guessing games: An experimental study. *The American Economic
492 Review* 96, 1737–1768.
- 493 Crawford, V.P., Costa-Gomes, M.A., Iriberri, N., 2013. Structural models
494 of nonequilibrium strategic thinking: Theory, evidence, and applications.
495 *Journal of Economic Literature* 51, 5–62.
- 496 Daw, N.D., Niv, Y., Dayan, P., 2005. Uncertainty-based competition between
497 prefrontal and dorsolateral striatal systems for behavioral control. *Nature
498 Neuroscience* 8, 1704–1711.

- 499 Efron, B., 1979. Bootstrap methods: Another look at the jackknife. The
500 Annals of Statistics 7, 1–26.
- 501 Elo, A.E., 1978. The Rating of Chess Players, Past and Present. Arco, New
502 York.
- 503 Embretson, S.E., Reise, S.P., 2000. Item Response Theory for Psychologists.
504 Lawrence Erlbaum Associates, Mahwah, NJ.
- 505 Ericsson, K.A., Krampe, R.T., Tesch-Römer, C., 1993. The role of deliberate
506 practice in the acquisition of expert performance. Psychological Review
507 100, 363–406.
- 508 Fischer, G.H., 1973. The linear logistic test model as an instrument in edu-
509 cational research. Acta Psychologica 37, 359–374.
- 510 Gigerenzer, G., Goldstein, D.G., 1996. Reasoning the fast and frugal way:
511 Models of bounded rationality. Psychological Review 103, 650–669.
- 512 Gobet, F., Simon, H.A., 1996. Templates in chess memory: A mechanism
513 for recalling several boards. Cognitive Psychology 31, 1–40.
- 514 Griffiths, T.L., Lieder, F., Goodman, N.D., 2015. Rational use of cognitive
515 resources: Levels of analysis between the computational and the algorithmic.
516 Topics in Cognitive Science 7, 217–229.
- 517 de Groot, A.D., 1965. Thought and Choice in Chess. Mouton, The Hague.
- 518 Huys, Q.J.M., Eshel, N., O’Nions, E., Sheridan, L., Dayan, P., Roiser, J.P.,
519 2012. Bonsai trees in your head: How the Pavlovian system sculpts goal-

520 directed choices by pruning decision trees. PLoS Computational Biology
521 8, e1002410. doi:[10.1371/journal.pcbi.1002410](https://doi.org/10.1371/journal.pcbi.1002410).

522 Kahneman, D., Tversky, A., 1979. Prospect theory: An analysis of decision
523 under risk. *Econometrica* 47, 263–291.

524 Lieder, F., Griffiths, T.L., 2020. Resource-rational analysis: Understanding
525 human cognition as the optimal use of limited computational resources.
526 *Behavioral and Brain Sciences* 43, e1.

527 McIlroy-Young, R., Sen, S., Kleinberg, J., Anderson, A., 2020. Aligning
528 superhuman AI with human behavior: Chess as a model system, in: Pro-
529 ceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery
530 and Data Mining (KDD), pp. 1677–1687. doi:[10.1145/3394486.3403219](https://doi.org/10.1145/3394486.3403219).

531 Monroe, D., Eilender, G., Chalmers, P., Tang, Z., Anderson, A., 2026. Chess-
532 former: A unified architecture for chess modeling, in: International Con-
533 ference on Learning Representations (ICLR). arXiv:2605.19091.

534 Nagel, R., 1995. Unraveling in guessing games: An experimental study. *The*
535 *American Economic Review* 85, 1313–1326.

536 Newell, A., Simon, H.A., 1972. *Human Problem Solving*. Prentice-Hall,
537 Englewood Cliffs, NJ.

538 Nosek, B.A., Ebersole, C.R., DeHaven, A.C., Mellor, D.T., 2018. The pre-
539 registration revolution. *Proceedings of the National Academy of Sciences*
540 115, 2600–2606.

- 541 van Opheusden, B., Kuperwajs, I., Galbiati, G., Bnaya, Z., Li, Y., Ma, W.J.,
542 2023. Expertise increases planning depth in human gameplay. *Nature* 618,
543 1000–1005. doi:[10.1038/s41586-023-06124-2](https://doi.org/10.1038/s41586-023-06124-2).
- 544 Rasch, G., 1960. Probabilistic Models for Some Intelligence and Attainment
545 Tests. Danish Institute for Educational Research, Copenhagen.
- 546 Ratcliff, R., McKoon, G., 2008. The diffusion decision model: Theory and
547 data for two-choice decision tasks. *Neural Computation* 20, 873–922.
- 548 Regan, K.W., Haworth, G.M., 2011. Intrinsic chess ratings, in: *Proceed-*
549 *ings of the 25th AAAI Conference on Artificial Intelligence*, pp. 834–839.
550 doi:[10.1609/aaai.v25i1.7951](https://doi.org/10.1609/aaai.v25i1.7951).
- 551 Ruoss, A., Delétang, G., Medapati, S., Grau-Moya, J., Wenliang, L.K., Catt,
552 E., Reid, J., Lewis, C.A., Veness, J., Genewein, T., 2024. Amortized plan-
553 ning with large-scale transformers: A case study on chess, in: *Advances*
554 *in Neural Information Processing Systems (NeurIPS)*, pp. 65765–65790.
555 arXiv:2402.04494.
- 556 Russell, S., Wefald, E., 1991. Principles of metareasoning. *Artificial Intelli-*
557 *gence* 49, 361–395.
- 558 Samejima, F., 1969. Estimation of latent ability using a response pattern of
559 graded scores. *Psychometrika Monograph Supplement* 34, 1–97. doi:[10.](https://doi.org/10.1007/BF03372160)
560 [1007/BF03372160](https://doi.org/10.1007/BF03372160).
- 561 Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A.,
562 Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., Lillicrap, T., Simonyan,

- 563 K., Hassabis, D., 2018. A general reinforcement learning algorithm that
564 masters chess, shogi, and Go through self-play. *Science* 362, 1140–1144.
- 565 Simon, H.A., 1955. A behavioral model of rational choice. *The Quarterly*
566 *Journal of Economics* 69, 99–118.
- 567 Simon, H.A., 1956. Rational choice and the structure of the environment.
568 *Psychological Review* 63, 129–138.
- 569 Stahl, D.O., Wilson, P.W., 1995. On players' models of other players: Theory
570 and experimental evidence. *Games and Economic Behavior* 10, 218–254.
- 571 Strzalecki, T., 2014. Depth of reasoning and higher order beliefs. *Journal of*
572 *Economic Behavior & Organization* 108, 108–122. doi:[10.1016/j.jebo.](https://doi.org/10.1016/j.jebo.2014.09.002)
573 [2014.09.002](https://doi.org/10.1016/j.jebo.2014.09.002).
- 574 Tversky, A., Kahneman, D., 1974. Judgment under uncertainty: Heuristics
575 and biases. *Science* 185, 1124–1131.